

Paritran

Auditable cyber-fraud investigation support on government-controlled hardware

From complaint to conviction

SOLUTION DOCUMENT

Category 2 student team | ID 10_338

PS-69EEFE4F8CD1C | Cyber Crime Branch, Ahmedabad City

Truth boundary

Working prototype measured on synthetic data. No real deployment, external institution validation, autonomous legal decision, or production admissibility claim is made.

Canonical claim source SHA-256

800091205ad1e7c2a22a7add37f8d8fcfa1e5094f8737806bea56f5229fbc357

1. Problem and scope

Detection, freezing, and registry systems are essential. Investigators still need to join related complaints, reconstruct the path of funds, map cited law, and preserve a reviewable custody record. Paritran addresses that operational handoff.

The product assists an officer. It does not name a person guilty, certify evidence, or replace legal review.

Build state

- Full prototype completed across ten milestones.
- 191 tests passing in the recorded build state.
- Public repository and GitHub Pages publication.
- Synthetic demonstration data only, with zero real PII.

Team

Ayush Tiwary: architecture, grounding, audit core, legal mapping, and on-premise build.

Aditya Arora: graph analytics, money trail, synthetic data, dashboard, and frontend.

2. System design

- Ingest: hash every input and attach source identity.
- Resolve: extract identifiers and join complaints by shared infrastructure.
- Link: surface candidate syndicate networks with graph methods.
- Trace: follow value through the directed ledger toward cash-out.
- Map: combine rules, BM25, and InLegalBERT; abstain on disagreement.
- Draft: produce cited language behind the F9 groundedness gate.
- Preserve: append every action to a tamper-evident custody chain.
- Review: require officer acceptance, rejection, and sign-off.

Design doctrine

The deterministic core never treats generated prose as evidence. The language layer may draft, but source checks and human review govern release.

3. Canonical measured results

All performance and scale results below are the File 2, Part B values. Network and linkage results use synthetic ground truth and must be described that way.

Measure	Result	Boundary
Linkage P / R / F1	0.957 / 0.966 / 0.962	Synthetic ground truth
Money trail	90.8 percent	Directed reachability
Legal mapping	100 percent, 8/8 and 15/15	Only when three paths agree
BM25 floor	52.4 percent	Condensed v1 corpus
F9 stub	40 / 10 / 0	Passed / withheld / leaked
F9 live gemma	5 / 1 / 0	Passed / withheld / leaked
NER P / R	0.74 / 1.0	Rule-augmented identifiers
Scale	297 complaints, 6/6 networks	Synthetic data
Custody	Verified and tamper-evident	Not immutable against full rewrite

Precision via abstention

The legal-mapping result is not a blanket accuracy claim. It is 100 percent only on the high-confidence subset where three independent paths agree: 8/8 and 15/15. All other cases route to a human officer.

4. Evidence, custody, and security

- Every input and output receives a SHA-256 digest.
- Custody is hash-chained and verified; silent edits break verification.
- A hash chain is tamper-evident, not magically immutable against a privileged full rewrite.
- The F9 gate withholds unsupported drafted claims.
- Role-based access and human sign-off preserve accountability.
- The demo is designed for local operation. Offline state is verified by the operator and environment, not inferred from marketing copy.
- Current Python dependency audit runs without advisory suppressions.

Legal boundary

Paritran can draft and pre-fill a Section 63 packet with citations and integrity material. Named custodians and qualified reviewers remain responsible for review and signature. Counsel-approved admissibility remains TODO-VERIFY.

5. Pilot and verification

A pilot should run on government-controlled hardware with an approved data-custody plan, an offline operating procedure, named roles, acceptance criteria, rollback, and a signed verification receipt.

Minimum acceptance checks

- Reproduce the canonical synthetic run without changing the frozen baseline.
- Verify linkage, money trail, F9, NER, and custody outputs against the receipt.
- Exercise role boundaries, export review, rejection paths, and recovery.
- Record network posture and physical controls at the venue.
- Do not introduce real complainant data until authority, minimisation, retention, and deletion rules are approved.

Known limitations

Real-institution validation, production data integration, counsel-approved certificate wording, production key custody, and third-party operational acceptance remain TODO-VERIFY.

Repository: github.com/ayushtiwary-ops/paritran

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